

Evaluating DPO’s Implicit Reward as a Preference Scorer Under Distribution Shift

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Abstract

Direct Preference Optimization (DPO) claims to skip the reward-model stage of RLHF entirely: a DPO-trained policy is itself, implicitly, a reward model. We test this claim as a preference-ranking task. Under a training budget matched between conditions, we train a Qwen2.5-0.5B policy with LoRA from one shared SFT checkpoint into three scorers — a DPO policy’s implicit reward, an explicitly trained Bradley–Terry reward model, and an SFT log-probability baseline with no reward-specific training — and evaluate all three as pairwise classifiers in distribution, on four RewardBench distribution-shift subsets, and on a cross-dataset shift set, across five seeds. In distribution, the explicit reward model wins (0.663 vs. 0.600 length-controlled accuracy, all five seed-level McNemar tests significant): **H1 is supported**. Out of distribution the predicted widening does not occur; where a resolvable difference exists (two of five shifted sets), it favors the implicit reward instead: **H2 is not supported**. On the hardest subsets, the SFT baseline beats both trained scorers. An extended four-epoch run and a per-checkpoint bias-drift analysis confirm the in-distribution ranking is not an artifact of stopping early, and implicate a length-preference confound in part of that gap.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) is the standard recipe for aligning a pre-trained language model with human preferences: collect pairwise comparisons between candidate responses, train an explicit reward model to predict which response people preferred, and then optimize the policy against that learned reward with reinforcement learning (typically PPO), subject to a KL penalty that keeps the policy close to a reference model [1, 2]. That pipeline needs a separately trained reward model and an unstable, hyperparameter-sensitive reinforcement-learning stage. Direct Preference Optimization (DPO) [3] was proposed to skip both, training a policy directly on preference pairs with a simple supervised-style loss while implicitly optimizing the same KL-regularized objective; its title makes the strong version of the claim, that a DPO-trained policy *is*, secretly, also a reward model. That reward is recoverable as $\hat{r}(x, y) = \beta \log[\pi_\theta(y | x) / \pi_{\text{ref}}(y | x)]$, but the equivalence is exact only at the optimum, which a trained policy is not guaranteed to reach; what we test here is $\hat{r}$ as a practical preference scorer, in and out of distribution, not the theorem itself.

Christiano et al. [1] introduced the core idea of learning a reward function from pairwise human preferences and optimizing a policy against it with reinforcement learning; its strength is generality (only comparative, not absolute, feedback is needed), but routing through PPO makes the resulting pipeline sensitive to reward-model quality and prone to reward hacking under prolonged optimization. Ouyang et al. [2] scaled this to language models with InstructGPT, explicitly training a Bradley–Terry-style reward model [4] and then fine-tuning the policy against it; the alignment gains were large, but the paper documents the same PPO instability and adds the cost of training

and serving a separate reward model. Gao et al. [5] quantify how far a learned reward can be optimized before proxy and true reward diverge — the failure our held-out comparison probes — though their scaling laws are fitted against synthetic gold reward models rather than human labels. DPO [3] removes that stage by training the policy directly on preference pairs with a simpler, more stable loss; its limitation, as detailed above, is that it discards the reward model as an explicit, separately-evaluable object, so any downstream reuse of $\hat{r}$ as a scorer has not been validated the way an explicit reward model would be, and we did not identify prior work that directly compares its preference-ranking performance against an explicitly trained reward model under matched training conditions. SimPO [6] is one of several post-DPO variants that drop the reference policy for a length-normalized reward, saving memory and curbing length bias, but in doing so it removes the very $\pi_\theta/\pi_{\text{ref}}$ ratio whose reuse as a reward model we test here. RewardBench [7] is the closest prior work to a stress test of reward models, built around adversarial, distribution-shifted comparisons, but it evaluates only explicitly-trained reward models, not policies read out as implicit ones. Two further ingredients make our question tractable at small scale: LoRA [8] allows several full training conditions to be run and compared on a single consumer GPU, at the cost of not testing full-parameter fine-tuning; and UltraFeedback [9] supplies the large, AI-scored preference pool our budget-matched training draws from, though its preferences reflect an AI judge’s biases rather than human raters’ directly.

This gap matters practically: if the implicit reward generalizes as well as an explicit one, DPO checkpoints could be reused as reward models, removing a training stage from the RLHF pipeline; if it does not, the “secretly a reward model” framing needs the qualification that it holds only in a narrow, training-distribution sense. We test two pre-registered hypotheses: **H1**, the explicit reward model beats the implicit reward on held-out in-distribution pairs, and **H2**, that gap widens under distribution shift. We treat a null result (implicit $\approx$ explicit) as equally publishable, since it would itself substantiate DPO’s claim empirically at small scale.

1.1 Our Contribution

- **Budget-matched pipeline:** the DPO policy and the explicit reward model differ only in their loss (Figure 1).
- **Correctness pass:** four silent bugs found and fixed, and the implicit-reward extraction verified against the DPO trainer’s own computation (§3.2).
- **Evaluation:** both scorers, plus an SFT log-probability baseline, scored as pairwise classifiers across six test sets over five seeds (§2.4).
- **Ablations:** a four-epoch extended-training run and a per-checkpoint length-bias-drift analysis (§3.7).

2 Material and Methods

2.1 Overview

Figure 1 shows the overall pipeline. A single supervised fine-tuning (SFT) pass produces the shared reference policy π_{ref} . Two training conditions then branch off from that identical starting point: DPO produces a policy π_θ from which the implicit reward is read off after training with no further reward-specific training, and a Bradley–Terry reward model is trained explicitly on the same pairs. Both are evaluated as pairwise preference classifiers on held-out data.

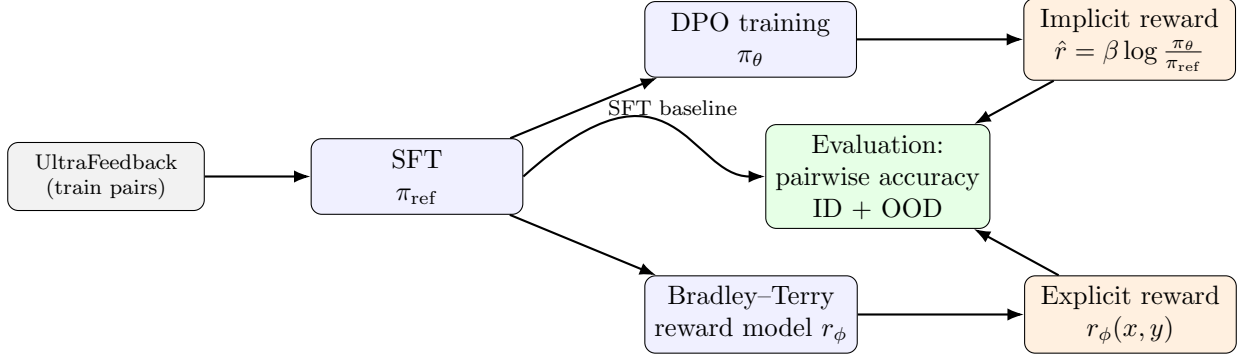


Figure 1: Pipeline overview. *Budget-matched* means both conditions share one SFT checkpoint, the same preference pairs, LoRA configuration, effective batch size, and optimizer steps; only the loss differs. π_{ref} is also scored directly, as the SFT log-probability baseline.

2.2 Dataset Description

All data are established public benchmarks (Table 1). UltraFeedback [9] supplies both the training pairs and the in-distribution held-out test set. RewardBench [7] supplies four distribution-shift subsets — Chat, Chat-Hard, Safety, and Reasoning — built specifically to stress-test reward models on harder and adversarially-constructed comparisons. Anthropic’s HH-RLHF (harmless-base split) supplies a cross-dataset shift test, since it comes from a different data-collection process entirely. Every example is normalized to the same (*prompt*, *chosen*, *rejected*) schema. Training pairs for the matched comparison are drawn from a common length-eligible pool so both conditions see the identical first 8,000 examples (Table 4 details why this pool is necessary).

Table 1: Evaluation sets. “Length-matched” counts the pairs surviving the 1.2 length-ratio filter (§2.4); the three marked † fall below 100 (§3.8).

Set	Role	Distribution	Pairs	Length-matched
UltraFeedback	training + ID test	in-distribution	1,997	442
RewardBench: Chat	OOD test	distribution shift	358	32 †
RewardBench: Chat-Hard	OOD test	distribution shift	456	62 †
RewardBench: Safety	OOD test	distribution shift	740	83 †
RewardBench: Reasoning	OOD test	distribution shift	1,431	908
HH-RLHF (harmless)	OOD test	cross-dataset shift	2,312	276

2.3 Tools, Models, and Algorithms Used

The base model for every condition is Qwen2.5-0.5B, a small, publicly available instruction-capable language model, chosen so that all three training conditions plus the extended-training ablation fit on a single 24 GB GPU (NVIDIA A10G). All fine-tuning uses Low-Rank Adaptation (LoRA) [8] with rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to every attention and MLP projection matrix, via HuggingFace’s TRL library. The shared reference policy π_{ref} is produced by one epoch of SFT on the *chosen* responses from 32,000 length-eligible UltraFeedback pairs (1,998 optimizer steps). (π_{ref} is a shared input to the matched comparison, not one of the compared conditions, so it is not budget-limited.) Figure 2 summarizes the three training objectives.

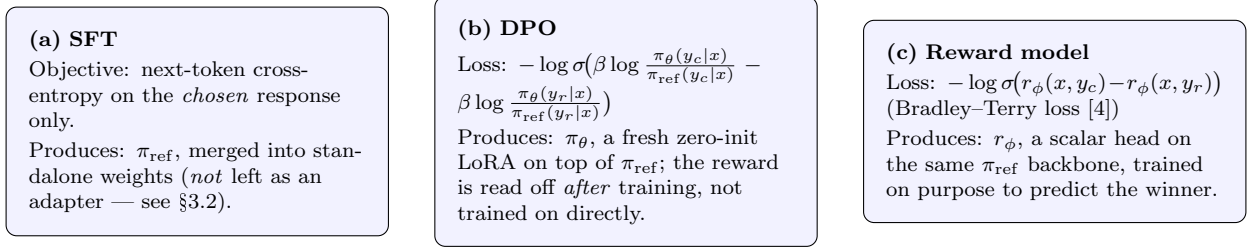


Figure 2: The three training objectives, all starting from the identical π_{ref} checkpoint produced by (a).

Table 2 lists the hyperparameters shared across SFT, DPO, and the reward model; the only quantities that differ between the two conditions are the loss function itself and, for the extended-training ablation, the number of epochs.

Table 2: Shared hyperparameters; $\beta = 0.1$ is DPO’s implicit-reward temperature.

Base model	Qwen2.5-0.5B
LoRA rank / α / dropout	16 / 32 / 0.05
LoRA target modules	q,k,v,o,gate,up,down projections
Effective batch size	16 (batch 4 $\times$ grad. accum. 4)
Learning rate (SFT / DPO & RM)	2×10^{-4} / 5×10^{-5} , cosine schedule
Precision	bf16 (single NVIDIA A10G)
Training pairs (headline run)	8,000, budget-matched, 1 epoch
DPO β (headline run)	0.1
Seeds (headline run)	5
Extended-training ablation	4 epochs, 2 seeds, held-out eval every 100 steps

2.4 Performance Evaluation

Every trained scorer produces a scalar $s(x, y)$: for the implicit reward, $s = \hat{r} = \beta \log[\pi_{\theta}(y | x)/\pi_{\text{ref}}(y | x)]$; for the reward model, $s = r_{\phi}(x, y)$; for the SFT baseline, $s = \log \pi_{\text{ref}}(y | x)$. Each s is used only for ranking — the scorer’s decision is the sign of $s(x, y_c) - s(x, y_r)$ — and *pairwise accuracy*, the fraction of pairs ranked correctly, is the primary outcome.

Length-controlled accuracy restricts this to pairs with $\max(L_c, L_r)/\max(1, \min(L_c, L_r)) \leq 1.2$, where L_c, L_r are character lengths of the response text (prompt excluded, un-tokenized); this filter is aggressive on the smaller RewardBench subsets. We also report non-neural floors (random, pick-longer, TF-IDF+LR) with no language-model fine-tuning, for comparison alongside the SFT baseline.

ECE is computed over 10 equal-width confidence bins, per seed on the full test set (not length-controlled) and averaged. *McNemar’s exact test* (a binomial test on the discordant pairs) is run separately per seed between the implicit and explicit scorers; we report the resulting range of five p-values rather than their mean, since independent p-values should not be averaged. The reported 95% CI is $\bar{x} \pm t_{0.975,4} s/\sqrt{5}$ over the five seed-level values of a given (test set, scorer, metric) triple, where s is their sample standard deviation (§3.8 discusses what this interval does and does not capture). We write *CI excludes/includes 0* for whether a paired explicit-minus-implicit difference is resolvable at this sample size.

3 Experimental Analysis

Code, configuration files, and the full pipeline used to produce every number and figure below are available at <https://github.com/silvererudite/dpo-implicit-reward>, with every per-pair score persisted to disk so every number below is recomputable without re-running any model.

3.1 Non-neural reference floors

Table 3 shows the zero-training reference floors. Pick-longer wins RewardBench-Chat (0.805) but collapses on Chat-Hard (0.294), which inverts that cue by construction, so raw accuracy on these benchmarks is substantially a length signal; this motivates length-controlled accuracy as the primary metric below. TF-IDF+LR lands below chance on Reasoning (0.282), suggesting the lexical cues it relies on are actively misleading on that subset.

Table 3: Non-neural reference floors (pairwise accuracy, no language-model training; TF-IDF+LR does fit a logistic regression). Pick-shorter is omitted as it is exactly $1 - \text{pick-longer}$.

Test set	Random	Pick-longer	TF-IDF + LR
UltraFeedback (ID)	0.500	0.568	0.618
RewardBench: Chat	0.500	0.805	0.754
RewardBench: Chat-Hard	0.500	0.294	0.382
RewardBench: Safety	0.500	0.420	0.450
RewardBench: Reasoning	0.500	0.397	0.282
HH-RLHF (harmless)	0.500	0.445	0.453

3.2 Pipeline correctness pass

We cross-checked our implicit-reward extraction against TRL’s own internal DPO reward computation on a batch of trained-on pairs, on five points: (i) our tokenization matches TRL’s token-for-token; (ii) our per-pair values agree with TRL’s logged ones up to a small residual attributable to batching and floating point (padded batched forward passes inside TRL versus our unpadded per-example ones); (iii) the extracted reward is exactly linear in β , ruling out a scaling bug; (iv) an unmasked-prompt control shifts the reward far more than that residual, ruling out a masking bug by comparison; (v) π_{ref} is bit-identical to TRL’s adapter-disabled reference. The residual’s exact source is not characterized, but it does not affect the reported metrics: pairwise accuracy depends only on the *sign* of each scorer’s own difference.

Bringing up the training pipeline separately surfaced four correctness bugs, none of which crashed — all would have produced plausible, wrong results (Table 4).

Table 4: Correctness bugs found and fixed before trusting any result. The first two fixes are why the SFT adapter is merged into standalone weights rather than left live beneath DPO/reward-model training.

Bug	Symptom if unfixed	Fix	Verified by
Wrong π_{ref}	DPO trains against the raw base model, not the SFT policy	Merge the SFT adapter into the base weights before DPO/RM	DPO loss starts at exactly $\ln 2$
Frozen RM head	Scalar head stays at random init; training does nothing to it	Merge SFT first, then attach a fresh sequence-classification head	<code>trainable == True</code> (measured)
Train-pair mismatch	RewardTrainer drops long pairs, DPOTrainer truncates them — 7,520 vs. 8,000 pairs at the same nominal budget	Draw both trainers from one common length-eligible pool	Pool hash-identical between runs
Multi-GPU batch inflation	HF Trainer auto-wraps in <code>DataParallel</code> ; effective batch silently $4\times$ larger	Pin every run to a single GPU	—

3.3 H1: in-distribution comparison

In distribution, the explicit reward model wins (Table 5) in every individual seed, with all five seed-level McNemar tests significant ($p < 0.0001$ – 0.0019). **H1 is supported.**

Table 5: Length-controlled pairwise accuracy (mean $\pm$ 95% CI, 5 seeds). “Diff.” is explicit minus implicit, paired within each seed.¹ “95% CI excludes/includes 0” is defined in §2.4; a negative, CI-excludes-0 difference means the implicit reward is ahead.

Test set	DPO implicit	Explicit RM	SFT baseline	Diff. (E–I)	95% CI
UltraFeedback (ID)	0.600 ± 0.026	0.663 ± 0.009	0.529	$+0.062 \pm 0.030$	excl. 0
RewardBench: Chat	0.800 ± 0.044	0.747 ± 0.080	0.500	-0.053 ± 0.058	incl. 0
RewardBench: Chat-Hard	0.535 ± 0.052	0.615 ± 0.061	0.661	$+0.079 \pm 0.109$	incl. 0
RewardBench: Safety	0.492 ± 0.036	0.549 ± 0.059	0.530	$+0.058 \pm 0.074$	incl. 0
RewardBench: Reasoning	0.850 ± 0.003	0.663 ± 0.019	0.906	-0.187 ± 0.020	excl. 0
HH-RLHF (harmless)	0.472 ± 0.016	0.429 ± 0.036	0.435	-0.043 ± 0.030	excl. 0

3.4 H2: does the gap widen out of distribution?

Pairing the explicit-minus-implicit difference *within* each seed, so run-to-run variance cancels, three of five OOD sets show no resolvable difference (Chat, Chat-Hard, Safety); on the two that do, Reasoning and HH-RLHF, the difference runs opposite to H2, favoring the implicit reward (Figure 3, Table 5). **H2 is not supported.**

¹Computed from unrounded per-seed values, so it can differ by up to 0.001 from the rounded means shown.

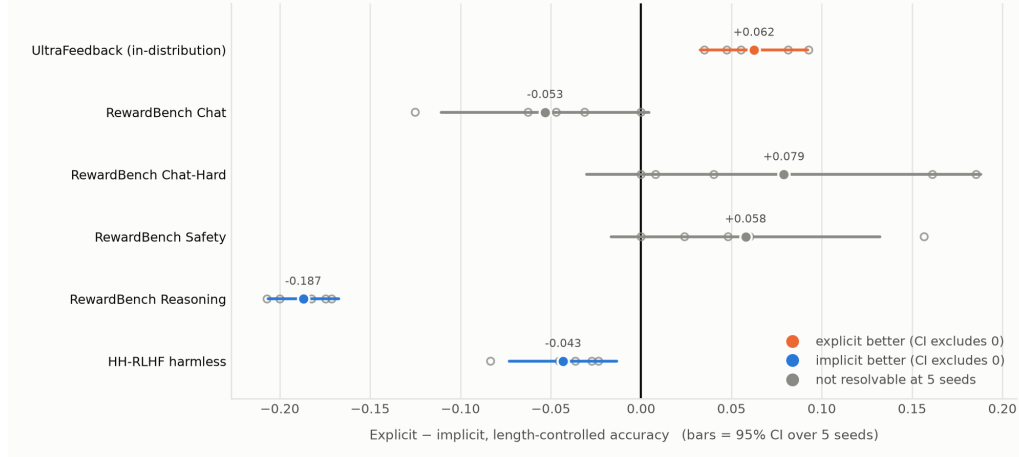


Figure 3: Explicit minus implicit length-controlled accuracy, paired within each seed; bars crossing zero (grey) are not resolvable.

3.5 An unplanned finding: reward training can underperform the SFT baseline

Scored against the SFT log-probability baseline instead of chance, the baseline beats *both* trained scorers on RewardBench-Reasoning; on Chat-Hard it beats the implicit reward resolvably, while the explicit model’s gap is not resolvable (Figure 4). Reward-model training, explicit or implicit, can therefore leave a scorer worse at ranking preferences than the SFT policy’s own untuned log-probability, on exactly the subsets RewardBench designed to be hard and adversarial.

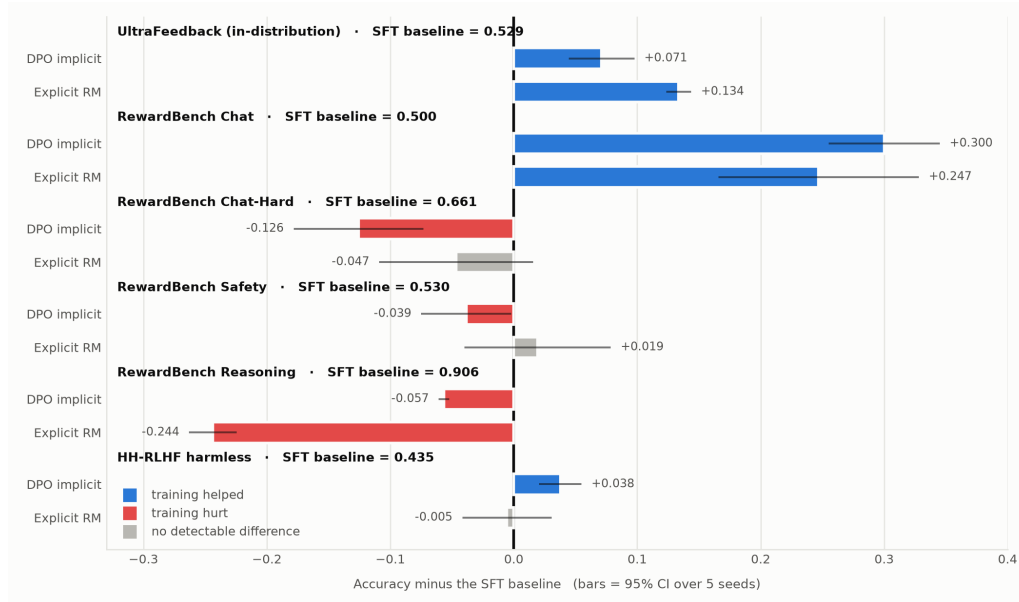


Figure 4: Accuracy minus the SFT log-probability baseline; bars left of zero mean reward training left the scorer *worse* at ranking.

3.6 Length confound and surface-feature associations

Length control matters: the explicit reward model scores below chance on Chat-Hard before length control (0.414) and recovers once length-matched (Table 5), a shift large enough to flip its ranking relative to chance. More generally, the explicit model and the SFT baseline are both associated with preferring longer, more formatted, more numeric responses relative to the human labels (10–16 points of deviation), while the implicit reward tracks the human rate closely (Figure 5). Response length is therefore one likely contributor to H1’s in-distribution gap and to the explicit model’s raw-accuracy collapse on length-inverted subsets like Chat-Hard.

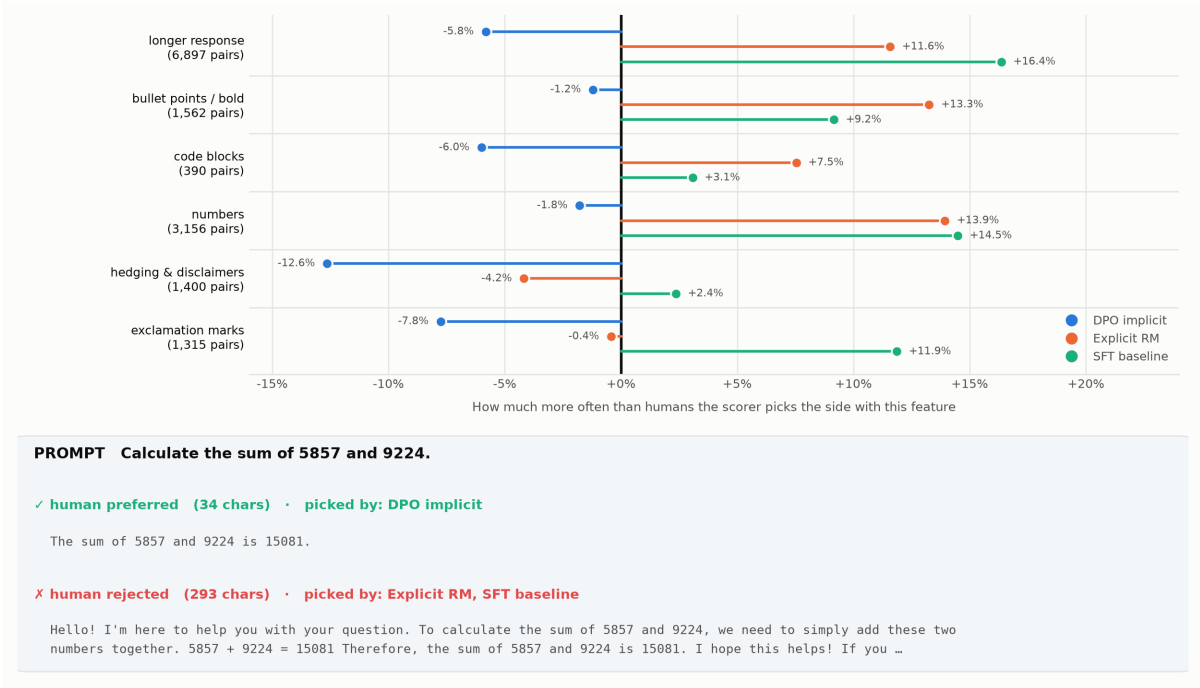


Figure 5: Deviation from the human rate on the same pairs, per surface feature (top), with one worked example (bottom).

3.7 Ablations: extended training and length-bias drift

Re-running both conditions for four epochs (two seeds), with held-out pairwise accuracy measured every 100 steps on a shared 400-pair slice, tests whether the one-epoch ranking is an artifact of stopping two still-improving models (Figure 6). The explicit reward model climbs to a peak of 0.731 near epoch 2 and then overfits; the implicit reward is flat from the start, peaking at 0.676 near epoch 1. The explicit model is ahead at every epoch past ~ 0.4 : its lead is ≈ 0.048 at the one-epoch mark (0.724 vs. 0.676) and 0.055 at each method’s own best checkpoint, slightly wider rather than narrower.² **H1 remains supported through the full four-epoch run; the ranking does not invert.** We stop short of calling this “convergence”: the reward model is still visibly overfitting, not plateaued, by epoch 4. Training loss keeps falling long after held-out loss turns (Figure 7), so it is not a usable stopping signal here; the reward model’s best checkpoint is near epoch 2, so four epochs is worse than one for it specifically.

²Different metric and sample from Table 5’s +0.062; not directly comparable.

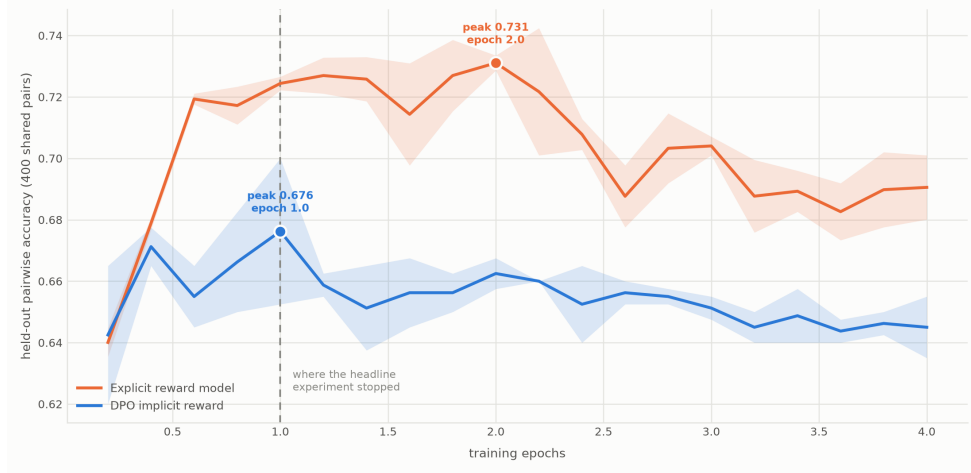


Figure 6: Held-out pairwise accuracy across four training epochs; the band spans two seeds.

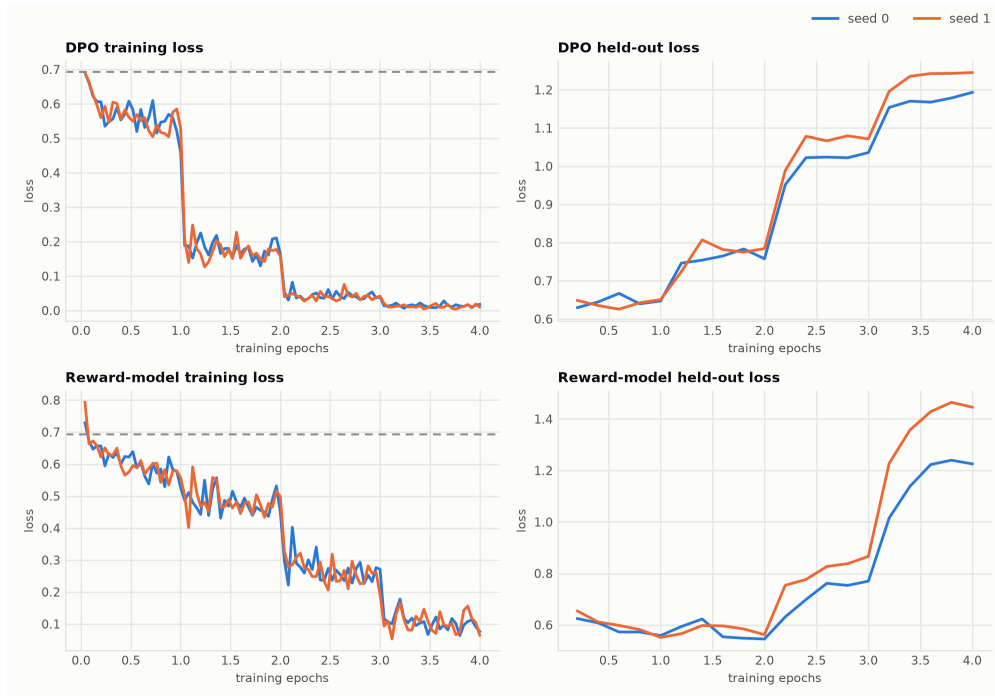


Figure 7: Training loss (left) against held-out loss (right) across the four-epoch run.

Scoring all eight saved checkpoints of each condition on 500 held-out pairs tests whether either scorer’s length bias grows with training (Figure 8; accuracy over the same checkpoints: Figure 6). The hypothesis is **wrong for the explicit model**: its length bias peaks early (+0.069 relative to the human rate at epoch 1.5) and fades to +0.002 by epoch 4, on roughly the same schedule as its accuracy peaking and falling. The implicit reward drifts the other way instead: its deviation from the human rate moves from -0.134 to -0.232 , so it ends up picking the *longer* response only 32.5% of the time against a human rate of 55.7%, with accuracy declining in step. This is consistent with an anti-length preference induced by DPO training that contributes to why the implicit reward saturates early and then degrades; both drifts are single-seed.

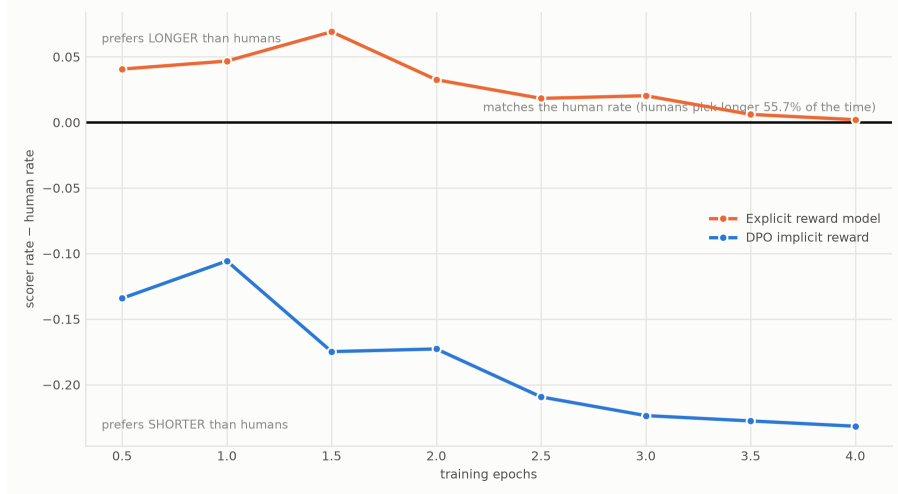


Figure 8: Length preference relative to the human rate, across eight checkpoints per model.

3.8 Threats to validity

Several limitations bound how far these results generalize.

Sample size. The three sets marked [†] in Table 1 fall below 100 length-matched pairs, the filter having removed up to roughly 90% of each. Their length-controlled numbers, reported to three decimals only for consistency with the rest of the table, indicate direction rather than precise estimates; a length-adjusted regression, rather than length filtering, would recover the lost statistical power.

What is not matched. Wall-clock time and total FLOPs are not matched, since DPO performs an extra frozen-reference forward pass every step and took approximately $1.7\times$ the reward model’s wall-clock time for the same optimizer steps. This is a matched *training budget* (Figure 1), not matched compute in the FLOP sense.

π_{ref} **was trained once.** Seeds vary only the second-stage (DPO/reward-model) training, so the reported intervals capture variance from that stage given one fixed reference, not variance of the method as a whole — which is also why the SFT baseline’s interval in Figure 4 has zero width.

Associations, not mechanisms. The surface-feature effects in Figure 5 are correlational and overlapping, since longer responses also tend to contain more numbers and bullets, so isolating length would need a joint model. Likewise the readings in §3.5 and §3.7 of *why* training degrades performance, or why the implicit reward drifts toward shorter responses, are consistent with the data rather than established causes.

Calibration is poor across the board (ECE 0.22–0.45 across scorers and sets), which matters if the practical pitch is reusing either kind of scorer as a probability estimate rather than only a ranker.

Scope. Every result is at one model scale (0.5B), one β (0.1), and one training budget (8,000 pairs); the $\beta \in \{0.05, 0.1, 0.3, 0.5\}$ and budget $\in \{2k, 8k, 32k\}$ ablations planned in the proposal remain future work.

4 Conclusion

Budget-matched from one shared starting checkpoint (Figure 1), an explicitly trained Bradley–Terry reward model beats DPO’s implicit reward on held-out, in-distribution preference pairs,

and the ranking survives an extended four-epoch run rather than reversing (**H1 supported**). The predicted widening of that gap under distribution shift does not occur; where a difference is resolvable, it favors the implicit reward instead (**H2 not supported**). Unplanned, an SFT log-probability baseline with no reward-specific training beats both trained scorers on the hardest, most adversarial subsets. DPO’s “secretly a reward model” claim therefore holds only in a narrow, in-distribution sense at this scale: an explicit reward model is measurably better in distribution, but under shift neither trained scorer consistently beats the untrained baseline, within the bounds of §3.8. Future work should run the planned β /budget sweeps and re-seed the reference policy itself so reported intervals describe the method rather than one initialization.

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